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KEY CONCEPTS IN CLINICAL EPIDEMIOLOGY

Estimands: what they are and why we should use them

Brennan C. Kahan[a,*], Declan Devane[b,c]

[a]MRC Clinical Trials Unit at UCL, Institute of Clinical Trials and Methodology, UCL, London, UK
[b]Health Research Board-Trials Methodology Research Network (HRB-TMRN), School of Nursing and Midwifery, University of Galway, Galway, Ireland
[c]Evidence Synthesis Ireland and Cochrane Ireland, School of Nursing and Midwifery, University of Galway, Galway, Ireland

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Abstract

In clinical trials, postrandomization events, such as treatment discontinuation or the use of rescue medication, can complicate the interpretation of results. An estimand is a precise description of the treatment effect that investigators wish to estimate. Estimands facilitate more straightforward interpretation of trial results by explicitly defining how postrandomization ''intercurrent'' events are incorporated into the research question. This article introduces the five key attributes of estimands (population, treatment conditions, endpoint, summary measure, and strategies for intercurrent events) and explains the five main strategies for managing intercurrent events (treatment policy, composite, while on treatment, hypothetical, and principal stratum). Using a practical example of a trial comparing cognitive behavioral therapy vs medication for mild anxiety, we demonstrate how different estimand choices lead to varying study designs, analyses, and interpretations. Understanding estimands helps researchers design better trials and enables stakeholders to determine if the results are relevant to their situation. We also explain how sensitivity analyses can be used to check the reliability of results by assessing how results change under different statistical assumptions. © 2025 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

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1. Background

Every clinical trial begins with a research question it aims to address, such as ''Is treatment with eight sessions of cognitive behavioral therapy (CBT) better than daily medication at reducing anxiety scores in people with mild anxiety?'' This research question guides us in designing the study, determining what data to collect, and analyzing those data [1−3]. However, sometimes unforeseen events occur during the study that we did not plan for or want to happen. For instance:

- Some participants assigned to CBT may not complete all eight sessions.
- Some participants assigned to medication may not take each dose they are supposed to.
- Some participants in the CBT group may start taking medication instead.

When these postrandomization events happen, they sometimes mean that the outcome data we gather from participants no longer addresses our research question [4,5]. For example, if we want to determine whether CBT is more effective than medication for mild anxiety, but some participants assigned to the CBT group immediately start taking medication instead, then the outcome data from these participants might not be relevant for assessing whether CBT is better than medication.

To determine whether outcome data following these types of postrandomization events are relevant for the research question, we need to formulate our question very precisely. For example, the question ''Is CBT more effective than medication for patients with mild anxiety?'' is too vague to clarify whether outcomes after receiving medication in the CBT group are relevant. To clarify this, we need to include the postrandomization event in our research question, for example, as:

- Is CBT with medication as indicated better than medication alone for patients with mild anxiety?
- Is CBT without medication better than medication alone for patients with mild anxiety?

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* Corresponding author. MRC Clinical Trials Unit at UCL, University College London, 90 High Holborn, London, WC1V 6LJ UK.
E-mail address: b.kahan@ucl.ac.uk (B.C. Kahan).

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In the first research question, outcomes following medication in the CBT group are relevant; for the second question, they are not.

Precisely defining the research question in this way also helps us understand how to run and analyze the trial. For example, for the second question, investigators would need to adjust the analysis to account for patients in the CBT group who started taking medication (eg, by using a statistical model to estimate what their outcomes would have been had they not started medication) [6−9].

Being clear about how postrandomization events are incorporated into the research question can also help stakeholders interpret and use study results. For instance, if a trial addresses the second question, this clarifies to patients interested in CBT, who may start on medication during their CBT, that the trial results might not be directly relevant to them.

The precise research question a study aims to answer is called an *estimand* [4,5,10−12]. Postrandomization events, such as initiating medication for patients in the CBT arm, are known as *intercurrent events*.

2. Why have estimands suddenly become more common?

While the importance of a clear research question has long been recognized, research questions have not always been clearly articulated [4,13,14]. A key issue was that research questions were often defined in terms of the trial's statistical analysis, notably with the terms *intention-to-treat* (ITT) and *per-protocol (PP)* [15]. However, these terms refer to the trial's analysis population (ie, the set of participants included in the analysis). Depending on how other parts of the analysis are conducted, they can be used to estimate quite different treatment effects [5,14]. For instance, an ITT analysis population can be used in conjunction with different statistical models to estimate the effect of an intervention regardless of nonadherence or if all patients had adhered (Box 1) [5,14]. Additionally, some analysis populations, such as PP analyses, do not always correspond to a clear research question [4,16].

This lack of clarity around a trial's exact research question was identified as a major barrier to the adequate appraisal of trials, and in 2019, the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use released the ICH E9(R1) addendum on estimands and sensitivity analyses, which sought to provide a coherent framework to clearly describe the research question of interest [4]. Below, we describe this framework and how it can inform a trial's design and analysis.

3. Key concepts and their application

An estimand is a precise definition of the treatment effect that investigators want to estimate (Box 1). It is distinct from the *estimator* (which is the statistical method used to analyze the data) and the treatment effect *estimate* (which is the numerical value computed from the statistical analysis). The three concepts can be thought of as follows: the *estimand* is the specific question, the *estimator* is the method by which data will be analyzed to answer that question, and the *estimate* is the actual answer obtained.

A clearly defined estimand, including how intercurrent events are considered, helps us to understand (i) what the exact research question being addressed is, and its relevance; and (ii) when each participant's outcome data are relevant and how it should be handled in the analysis to address the research question.

A common framework for defining estimands involves describing five key attributes [4,5]. Completing these five attributes ensures that all relevant parts of the research question are outlined, leaving no room for ambiguity.

The five attributes are listed in Table 1. In brief, they are:

1. Population
2. Treatment conditions
3. Endpoint/outcome variable
4. Summary measure
5. Strategies for managing intercurrent events

These attributes closely resemble those used in the PICO (Population, Intervention, Comparator, Outcome) framework, but they include two additional attributes to more accurately define the research question. The first additional attribute is the "summary measure," which describes how outcomes are compared between treatments (eg, using a risk difference, odds ratio, difference in means, etc.). The second additional attribute relates to how intercurrent events are incorporated into the research question, which is explained further below.

4. Intercurrent events

Intercurrent events are postrandomization occurrences that affect either the interpretation or existence of a participant's outcome [4,5]. When participants experience intercurrent events, this can meant that their outcome data are no longer relevant for answering the research question.

Intercurrent events usually refer to changes in treatment that occur during the course of the trial but can also include events such as death that affect the very existence of the outcome (see below). Common examples include:

- Stopping the assigned treatment early
- Not starting the assigned treatment
- Switching from one treatment arm to another
- Receiving nontrial medication (eg, rescue medication)

Intercurrent events can also refer to situations in which the outcome is no longer clearly defined after the event

Box 1 Key concepts.

Key definitions

Estimand: A precise description of the treatment effect to be estimated.

Estimator: The statistical method used to analyze the data and compute a treatment effect estimate.

Estimate: The numerical value computed from the estimator.

Intercurrent event: A postrandomization event which affects the interpretation or existence of the endpoint/outcome (common examples include treatment discontinuation, treatment switching, use of rescue medication, or death)

Links to other terminology and concepts

Intention-to-treat (ITT)/per-protocol (PP): The terms ITT and PP describe the analysis population that is used as part of the *estimator*, eg, ITT typically means that all randomized participants are included in the analysis according to their assigned treatment arm. Both ITT and PP analysis populations can be used to estimate different estimands, depending on which statistical model they are combined with. For example, an ITT analysis population can be used with a standard regression model to estimate a treatment policy estimand or with an instrumental variable approach to estimate a principal stratum estimand.

Pragmatic/explanatory trials: These terms refer to the specific objectives of investigators in carrying out a trial. A pragmatic trial is one intended to help understand how well an intervention works in "real-world" conditions, while an explanatory trial helps to understand how well an intervention works under "ideal" conditions. For both pragmatic and explanatory trials, there could be different estimands that reflect their overall objectives.

Protocol deviation: Protocol deviations are deviations from the trial protocol, such as enrollment of ineligible participants, or providing someone with the incorrect treatment. Some protocol deviations may also be intercurrent events (eg, receiving the incorrect treatment), but not all are. Similarly, some intercurrent events may not be protocol deviations (eg, receipt of rescue medication may be allowed by the protocol but still reflects an intercurrent event).

Example to illustrate concepts

Investigators design a randomized trial because they want to answer the question, "Are eight sessions of CBT more effective than daily medication at reducing anxiety?" They know that some patients in the CBT group may begin taking medication (an *intercurrent event*), so they must decide how this should be handled in their *estimand*, eg, do they want to know (i) the effect of CBT without medication vs medication alone; or (ii) the effect of CBT with medication as indicated vs medication alone?

Investigators decide they want to run a *pragmatic* trial to understand effect of CBT vs daily medication if used in real-world circumstances. Therefore, they decide to specify the use of a treatment policy strategy (ie, they are interested in the effect of CBT vs medication, regardless of whether patients in the CBT group begin taking medication), as this estimand most closely reflects the investigators' objective.

Investigators specify that their *estimator* will use an *ITT* analysis population (ie, include all randomized participants in their assigned treatment group) in conjunction with a linear regression model. After running the trial and collecting the data, they run their estimator and calculate their treatment effect *estimate* as a difference in means of −1.5 points on the anxiety scale (95% confidence interval −2.7 to −0.3).

occurs. The most common example is death. For instance, if the outcome is an anxiety score at 12 months, and a participant dies at month 5, then their anxiety score at month 12 does not exist. This differs from missing outcome data, which suggests that the outcome could have been measured but was not (see Box 2) [4,5].

As mentioned earlier, it is vital to include intercurrent events in the research question. There are five common methods, known as *strategies*, which are detailed in Table 2. Each strategy has different implications for whether outcome data remain relevant after the intercurrent event and how the study should be analyzed.

5. How to estimate different intercurrent event strategies?

The following section provides technical details on estimation approaches for different strategies. Readers who primarily interested in understanding estimand concepts may wish to focus on the practical example in Table 3 before returning to these details.

Different intercurrent event strategies will typically require different analysis populations, data, and statistical models. For some strategies, numerous different estimators can be used.

A treatment policy strategy can be estimated using an ITT analysis population (ie, including all randomized participants analyzed according to their randomized treatment assignment) in conjunction with a standard statistical model (eg, linear or logistic regression) applied to the

Table 1. The five attributes of an estimand and what they mean

Attribute	Definition	What this means
Population	Which patients are targeted by the research question	This describes which population of patients' study investigators want to estimate the treatment effect (eg, patients with mild anxiety)
Treatment conditions	Which treatments are of interest for the comparison	This describes which versions of treatment investigators want to compare (eg, eight sessions of CBT without the use of anti-anxiety medication vs anti-anxiety medication alone)
Endpoint/outcome variable	Which outcome is to be obtained for each individual to address the research question	Which outcome is used to compare treatments (eg, HADS Anxiety subscale at month 12)
Summary measure	How endpoints will be summarized and compared between treatments conditions (eg, risk ratio, odds ratio, etc)	Which method will be used to compare outcomes between treatments (eg, difference in means)
Strategies to handle intercurrent events	How each intercurrent event is included in the treatment effect definition	When postrandomization events occur, such as if some patients stop attending CBT early, whether investigators are interested in the patients' outcomes regardless of whether they stopped early, in what the patients' outcomes would be if they had not stopped early, etc

CBT, cognitive behavioral therapy; HADS, Hospital Anxiety and Depression Scale.

outcome data (where, importantly, all outcome data are used regardless of whether participants experienced an intercurrent event) [5].

Composite and while-on-treatment strategies can also be estimated using ITT analysis populations in conjunction with a standard statistical model; however, the outcome data are modified first before analysis. For a composite strategy, the outcome is modified by creating a composite outcome (eg, by assigning participants who experience the intercurrent event an unfavorable outcome value), and for a while-on-treatment strategy, the outcome is modified to only use data up to the point the intercurrent event occurs (eg, by redefining the outcome to be the average anxiety score before treatment discontinuation or 12 months) [5]. These approaches require high-quality data collection on both the outcome and the occurrence of the intercurrent event.

There are several ways to estimate hypothetical strategies. A common method is to assume that the outcomes of all randomized participants after the intercurrent event are not part of the target. Then, models such as inverse probability weighting, multiple imputation, or likelihood-based models (such as mixed models for repeated measures) are used to estimate what the outcomes would have been in the absence of the intercurrent event [6–9]. These methods are based on assumptions that cannot be tested about variables that affect both the intercurrent event and the outcome. They also need careful data collection on the outcome, the intercurrent event, and related variables before and after randomization.

A range of methods exists for estimating principal stratum strategies, involving various analysis populations and statistical models [17–19]. These methods typically require high-quality data collection on the outcome and the occurrence of the intercurrent event. Some methods may also require additional data collection on other baseline or postbaseline variables.

More thorough overviews of different estimation methods are available [5–9,17–20].

6. Sensitivity analyses to address analysis assumptions

Estimation methods for all intercurrent event strategies can rely on untestable assumptions to provide valid results. For example, if a hypothetical strategy estimator assumes that participants who discontinued treatment would have had similar outcomes to those who continued and had the same prognostic covariates (a "missing at random" assumption), sensitivity analyses might test what would happen if discontinuers had systematically worse outcomes (using methods such as delta-adjusted multiple imputation to shift their imputed values downward).

Assumptions commonly need to be made about the nature of any missing outcome data (which can affect all strategies),

Box 2 Missing outcome data and intercurrent events.

Missing outcome data are not an intercurrent event because it does not change whether the outcome is relevant for the research question.

For example, if a patient assigned to CBT begins taking medication (intercurrent event), their outcome may no longer be relevant for a "CBT without medication" estimand. In contrast, if that same patient simply doesn't return their questionnaire (missing data), this doesn't change whether their outcome would be relevant, only whether it is observed.

However, intercurrent events can lead to missing data, and both issues need to be addressed. If a patient discontinues CBT (intercurrent event) and then does not provide outcome data (missing data), both need to be considered: first, how treatment discontinuation is handled in the estimand, and second, how to address the missing outcome data in the analysis.

and principal stratum and hypothetical strategies will require additional assumptions, such as "no unmeasured confounding", monotonicity, or the exclusion restriction (described in detail elsewhere [8,9,17,18,21]). Because these assumptions cannot be verified from the data, it is important to perform additional "sensitivity" analyses to check the consistency of results under different assumptions while still aiming to answer the same research question. For example, when outcome data are missing, it is important to test how results change under different assumptions about the missing data using methods such as pattern-mixture models, delta-adjusted multiple imputation, or alternative approaches. These analyses help readers assess the robustness of the findings.

7. Practical example

Table 3 displays three possible estimands for a trial comparing CBT and medication for patients with mild anxiety. Each estimand uses different intercurrent event strategies (including treatment policy, hypothetical, and principal stratum, respectively). The first estimand addresses a pragmatic objective (ie, the effect of the intervention when used as part of routine practice [22]), whereas the latter two address explanatory objectives (ie, the effect of the intervention if implemented as intended [22]). The hypothetical estimand estimates the effect if all participants had fully adhered to their assigned treatment, while the principal

Table 2. Strategies for handling intercurrent events and implications for the trial

Name of strategy	Description	What it means for the trial
Treatment policy	Outcome data are relevant regardless of whether the event occurs	Investigators would use patient outcomes in the analysis regardless of whether they were affected by the intercurrent event
Composite	Outcome data for patients with the event are not relevant[a]; the event itself is relevant, so a composite outcome that includes the event is used.	Investigators would change the outcome for patients affected by the event (eg, by setting outcomes to a "failure" or a low score)
While on treatment	Outcome data up to the point of the event are relevant, but outcome data after the event are not	Investigators would change the outcome for patients who had the event to only include outcome data before the event (eg, by defining the outcome to whether patients had a clinical event up to 28 d or before the intercurrent event, whichever is first)
Hypothetical	Outcome data after the event are not relevant; investigators want to know what the outcome would have been if patients had not been affected by the event	Investigators would typically use a statistical model to estimate what patient outcomes would have been if they had not been affected by the intercurrent event.
Principal stratum	Outcome data for patients with the event are not relevant; only outcome data for patients who would not experience the event are relevant	Investigators would use a statistical model to try and identify which patients would not be affected by the event (eg, those who would adhere under either treatment assignment), then compare outcomes within the subpopulation who would not be affected by the event.

[a] Most composite outcomes do not use outcome data from patients who experience the postrandomiation event and instead assign patients with the event a particular outcome score (eg, assigning patients a "failure" status). However, some types of composites still include outcome data from patients who experienced the event (eg, hierarchical composite outcomes, such as those based on a win-ratio).

Table 3. Example of estimands for a trial comparing CBT vs medication. In practice, only a single estimand may be necessary to address the trial objectives (though >1 estimand could be specified); however, we present three estimands here to illustrate the different possibilities and the impact they have on the design and analysis of the trial

	Estimand 1 – Treatment policy estimand (effect of CBT vs medication regardless of nonadherence)	**Estimand 2 – Hypothetical estimand (effect of CBT vs medication if all patients fully adhered)**	**Estimand 3 – Principal stratum estimand (effect of CBT vs medication in those patients who adhere to their assigned treatment)**
Estimand			
Population	Patients with mild anxiety	Patients with mild anxiety	Patients with mild anxiety who would adhere to their assigned treatment (ie, attend at least 50% of CBT sessions if assigned to CBT and not receive CBT if assigned to medication alone)
Treatment conditions	Eight sessions of CBT with anti-anxiety medication as indicated vs medication alone, regardless of whether patients complete all sessions of CBT	Eight sessions of CBT without any anti-anxiety medication vs medication alone, targeting the scenario where all participants fully adhere	Eight sessions of CBT with anti-anxiety medication as indicated vs medication alone
Endpoint/variable	12-month HADS anxiety score	12-month HADS anxiety score	12-month HADS anxiety score
Summary measure	Difference in means[a]	Difference in means[a]	Difference in means[a]
Intercurrent events			
Stopping CBT early	Treatment policy strategy (effect regardless of whether some patients stop CBT early)	Hypothetical strategy (effect if all patients completed all eight sessions of CBT)	Principal stratum strategy (effect in those patients who adhere to both treatment assignments, ie, would attend ≥4 CBT sessions if assigned to CBT and not receive CBT if assigned to medication alone)
Taking anti-anxiety medication in the CBT arm	Treatment policy strategy (effect regardless of whether some patients in the CBT arm begin taking medication)	Hypothetical strategy (effect if no patients in the CBT arm began taking medication)	Treatment policy strategy (effect regardless of whether some patients in the CBT arm begin taking medication)
Estimand summary	The estimand is the difference in means for the 12-month HADS anxiety score between eight sessions of CBT with anti-anxiety medication as indicated vs medication alone, regardless of whether patients complete all sessions of CBT, in patients with mild anxiety.	The estimand is the difference in means for the 12-month HADS anxiety score between eight sessions of CBT without any anti-anxiety medication vs medication alone, if patients completed all sessions of CBT, in patients with mild anxiety.	The estimand is the difference in means for the 12-month HADS anxiety score between eight sessions of CBT with anti-anxiety medication as indicated vs medication alone, in patients with mild anxiety who would adhere to their assigned treatment (ie, attend at least 50% of CBT sessions if assigned to CBT and not receive CBT if assigned to medication alone).
Implications for the trial			
Design implications	Methods to improve adherence beyond what would be available in usual practice should not be used	Methods to improve adherence could be used (eg, additional support or reminders to attend CBT sessions, etc)	Methods to improve adherence to CBT beyond what would be available in usual practice would not typically be necessary (as the aim is to evaluate the effect of CBT in those who would adhere)
Analysis implications	Analysis could be done by including all randomized patients in the analysis according to their assigned treatment arm (ITT analysis population), in conjunction with a linear regression model	Analysis could be done by including all randomized patients in the analysis according to their assigned treatment arm, but censoring outcome data at the point an intercurrent event occurred. A linear regression model used with a method such as inverse probability weighting could be	Analysis could be done by including all randomized patients in the analysis according to their assigned treatment arm (ITT analysis population), in conjunction with a linear regression model using instrumental variables (two stage least squares) [17,18]

(*Continued*)

Table 3. Continued

	Estimand 1 – Treatment policy estimand (effect of CBT vs medication regardless of nonadherence)	Estimand 2 – Hypothetical estimand (effect of CBT vs medication if all patients fully adhered)	Estimand 3 – Principal stratum estimand (effect of CBT vs medication in those patients who adhere to their assigned treatment)
		used to re-weight the data to reflect what patient's outcomes would be if they had not stopped CBT early/started medication in the CBT group [6,8,9].	

[a] The summary measure is the difference in the average 12-month HADS anxiety score between treatment conditions. This could be estimated using adjusted or unadjusted models (eg, models that may adjust for baseline covariates such as age, sex, baseline anxiety, etc.). Adjusted models can improve statistical precision, but in this example, whether the model is adjusted or unadjusted does not change the target summary measure or estimand.

stratum estimand estimates the effect in the subpopulation that would adhere to their assigned treatment.

Each estimand requires different study methods and yields different information for end-users. Notably, both estimands 1 (treatment policy) and 3 (principal stratum) use an ITT analysis population but employ different statistical models, whereas estimand 2 (hypothetical) includes all randomized patients, similar to ITT, but censors their outcome data after nonadherence. This highlights a key difference between ITT and PP analyses vs treatment policy and other estimands — ITT and PP related to analysis principles (ie, methods used to address the research question), while estimand strategies define the target research question (Box 1).

In many trials, a single estimand will be sufficient to address the main objectives. For example, in a pragmatic trial evaluating CBT vs medication in real-world settings, a single estimand using a treatment policy strategy would address this. However, some trials may have multiple objectives, eg, to understand the effect of CBT both in real-world settings and under ideal conditions. This may be because investigators wish to address more than one objective or because different stakeholders (such as regulators and health technology agencies) may have different information needs. In these cases, multiple estimands can be defined, though it would be important to clarify which would be considered the primary estimand and how different estimands would be used as the basis for decision-making by different stakeholders.

The chosen estimand(s) can be written in trial protocols and Statistical Analysis Plans either using a ''table'' format (ie, listing the five estimand attributes in a table, as shown in Table 3) or using a narrative description (similar to the ''Estimand summary'' row in Table 3).

8. Notes and practical tips

Common pitfalls

- Confusing intercurrent events with missing data (see Box 2)
- Choosing intercurrent event strategies that don't align with the research question or stakeholder needs
- Not prespecifying in the trial protocol how intercurrent events will be handled
- Failing to collect adequate data on intercurrent events and variables needed for the planned analysis

Practical guidance

- Define estimands during the protocol development stage
- Consider what intercurrent events are likely to occur
- Choose strategies that match what stakeholders want to know
- Ensure the statistical analysis plan aligns with the estimand

Make sure to

- Discuss the choice of estimand(s) with key stakeholders, such as patients, health-care professionals, and regulatory bodies or health technology agencies as appropriate, to understand which estimands are most relevant for decision-making [5].
- Collect and report data on intercurrent events to help stakeholders better understand trial results [23,24].
- Report the chosen estimand(s), intercurrent event strategies, and related analyses so that readers can assess applicability and robustness

9. Conclusion

Clinical trials should answer the questions that key stakeholders, including patients and clinicians, want answered. Estimands provide a structured framework for precisely defining research questions in clinical trials. By explicitly considering how postrandomization events are incorporated into the research question, estimands help ensure that trial design, conduct, and analysis are aligned with the underlying scientific question. This clarity benefits both researchers in designing appropriate trials and

stakeholders in understanding the applicability of trial results to their specific contexts.

CRediT authorship contribution statement

Brennan C. Kahan: Writing − review & editing, Writing − original draft, Conceptualization. **Declan Devane:** Writing − review & editing, Conceptualization.

Declaration of competing interest

The authors have no competing interests to declare.

Data availability

No data was used for the research described in the article.

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Suggestions for further reading

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[2] Kahan BC, Hindley J, Edwards M, Cro S, Morris TP. The estimands framework: a primer on the ICH E9(R1) addendum. BMJ 2024;384:e076316.
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[4] De Silva AP, Leslie K, Braat S, Grobler AC. Application of the Estimand Framework to Anesthesia Trials. Anesthesiology 2024;141(1):13−23. *Content:* Describes examples of how to choose an estimand for a specific clinical area (anesthesia).
[5] Cro S, Kahan BC, Patel A, Henley A, C J, Hellyer P, et al. Starting a conversation about estimands with public partners involved in clinical trials: a co-developed tool. Trials 2023;24:443. *Content:* Includes a co-developed tool to help facilitate discussions with public partners about estimands.